WR Churn Risk Prediction Using a Synthetic Mobile MOBA Dataset

A portfolio-safe machine learning workflow for early retention risk modeling and reactivation-oriented player analysis

1. Executive Summary

This project demonstrates a churn-risk modeling workflow using a synthetic dataset designed to approximate player lifecycle patterns in a fast-paced mobile MOBA environment. The analysis focuses on early retention risk prediction by comparing Random Forest, XGBoost, and LightGBM on a Wild Rift-inspired tabular dataset.

Each row represents one player over a trailing 7-day behavioral window. The target variable, churn_risk, is a synthetic short-term retention risk label constructed from recency, activity, progression, social play, and friction-related signals. The goal is not to reproduce internal production logic, but to show an end-to-end modeling workflow that is suitable for portfolio presentation.

Across the evaluated models, XGBoost produced the strongest overall performance. It achieved 0.9927 AUC, 0.9637 accuracy, 0.9346 precision, 0.8979 recall, and 0.9158 F1-score in cross-validation. LightGBM remained competitive and produced the highest recall at 0.9429, making it relevant for workflows where minimizing missed higher-risk players is more important than maximizing precision.

From a business perspective, this workflow is most relevant for early retention monitoring and can also support reactivation targeting for players showing signs of declining engagement. The project should be interpreted as a synthetic churn-risk modeling showcase rather than a production churn system.

2. Problem Context

Player retention is a critical challenge in live-service games, especially in competitive mobile titles where progression pace, match frequency, social play, and frustration signals can all influence continued engagement. In a mobile MOBA setting, many early warning signals emerge before a player fully disengages. These include lower recent activity, widening login gaps, weaker progression, and reduced interaction with friends or premade groups.

The ability to identify elevated churn risk earlier creates practical opportunities for intervention. Product and operations teams can use risk monitoring to support onboarding improvements, targeted retention messaging, event participation nudges, or reactivation campaigns for players showing signs of drop-off.

This project frames the problem as short-term retention risk prediction rather than direct production churn estimation. That distinction is important because the dataset used here is synthetic and the target label is engineered for demonstration purposes. The objective is therefore to evaluate modeling logic, feature structure, and interpretation workflow in a portfolio-safe environment.

3. Dataset Design

The dataset used in this project is synthetic and was designed to approximate realistic player lifecycle patterns in a fast-paced mobile MOBA environment. It does not contain real Wild Rift production data and should not be interpreted as internal Riot Games telemetry.

The final dataset contains 40,034 rows and 27 columns, where each row represents one player over a trailing 7-day behavioral window. Regional coverage is limited to SEA-inspired player segments, including Singapore, Malaysia, the Philippines, Indonesia, Thailand, and Vietnam. This structure supports a retention analysis context while remaining safe for public portfolio use.

The feature set includes player lifecycle, engagement, progression, social play, friction, and monetization-related variables. Examples include matches_last_7d, active_days_last_7d, last_login_gap_days, ranked_match_ratio, win_rate_last_10, party_rate, report_count_30d, afk_count_30d, and spend_tier. Together, these variables create a more believable retention-risk modeling environment than a generic gaming behavior dataset.

The target variable, churn_risk, is a synthetic binary label intended to approximate elevated short-term disengagement risk. It is generated from a weighted combination of behavioral recency, activity intensity, progression, social play, and friction-related indicators. A positive label indicates higher retention risk rather than an observed churn event.

Table 1. Dataset Summary

Metric	Value
Rows	40,034
Columns	27
Unit of analysis	One player over trailing 7-day window
Regions	SG, MY, PH, ID, TH, VN
Positive class	churn_risk
Positive class rate	22.0%

4. Modeling Approach

The modeling workflow follows a standard supervised machine learning pipeline for tabular classification. The identifier column is excluded from model input, numeric features are median-imputed, and categorical features are imputed using the most frequent value before one-hot encoding. This preprocessing design ensures that mixed feature types can be handled consistently across all models.

Three tree-based algorithms are compared in this study: Random Forest, XGBoost, and LightGBM. Random Forest is included as a strong and interpretable baseline. XGBoost is used as a high-performing boosted-tree benchmark. LightGBM is included because it is efficient, well suited to tabular modeling, and commonly effective in retention-style prediction problems.

Model performance is evaluated using stratified 10-fold cross-validation. The evaluation metrics include accuracy, precision, recall, F1-score, and AUC. This metric set was chosen to balance overall discrimination quality with practical risk detection performance. In a retention context, recall is especially important because failing to identify a genuinely higher-risk player can be more costly than flagging a manageable number of lower-risk players.

5. Exploratory Observations

Initial exploratory checks suggest that the synthetic dataset captures plausible lifecycle variation for a mobile MOBA retention setting. The target remains imbalanced enough to resemble a realistic operational classification setting without becoming too sparse for demonstration purposes.

The transformed dataset also supports meaningful behavioral segmentation. Recent match activity spans a wide range of player states, from very low recent participation to highly active cohorts. This is useful because a retention-risk workflow should be able to distinguish between declining players and players who remain consistently engaged.

Regional comparisons also show moderate variation in synthetic churn-risk rates across SEA-inspired segments. Vietnam has the highest synthetic risk rate at 22.9%, followed by Singapore at 22.3%, Indonesia at 22.2%, Malaysia at 22.1%, Thailand at 21.7%, and the Philippines at 21.2%. These differences should not be interpreted as real market performance. They are included to create a more realistic cohort-analysis environment for portfolio use.

Figure 1. *Class Balance of Synthetic Churn Risk*

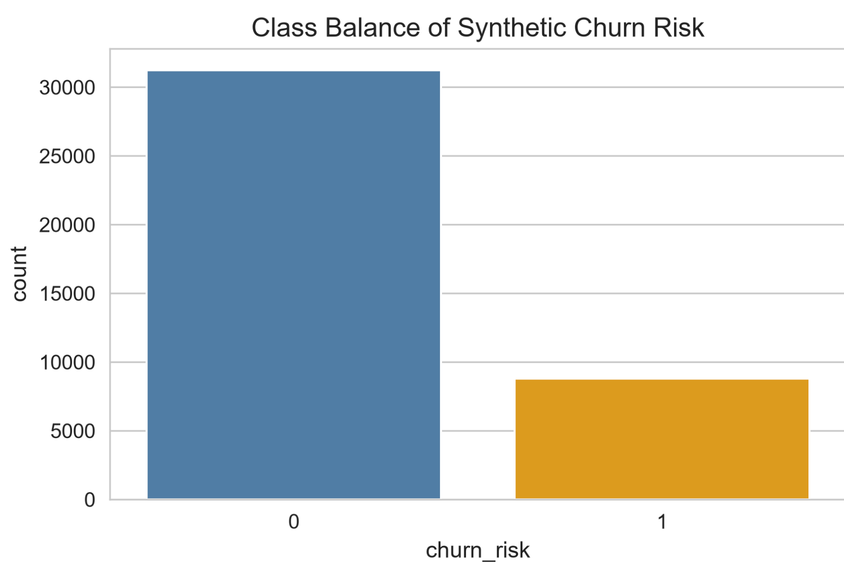

Figure 1 shows the class distribution of the synthetic churn_risk target. Higher-risk players account for 22.0% of the dataset, creating a realistic but manageable class imbalance for retention-risk modeling.

Figure 2. Distribution of Matches in Last 7 Days

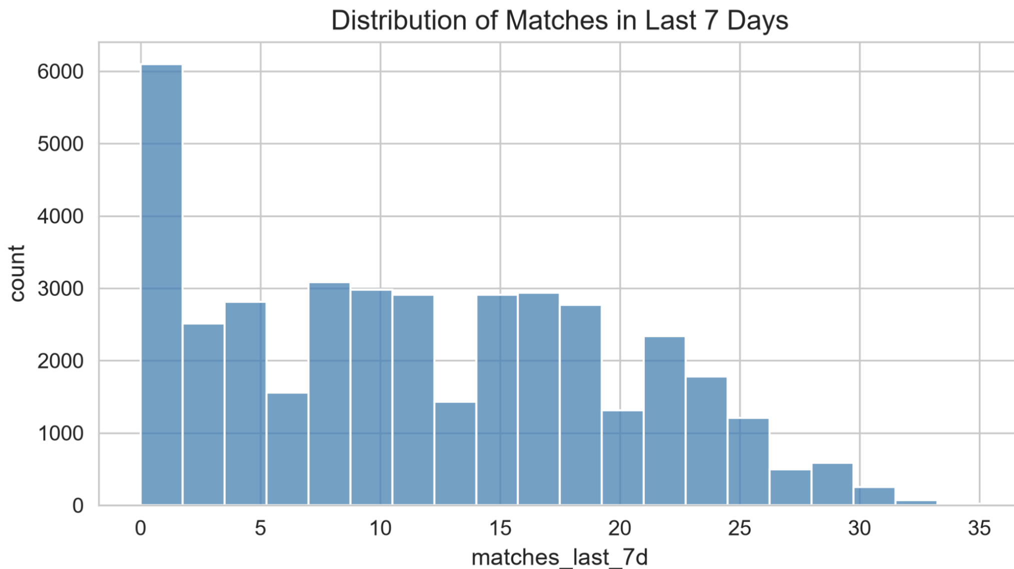

Figure 2 shows the distribution of recent match participation across the synthetic player base. This variable becomes the strongest predictor in the final model and acts as the most direct behavioral signal of short-term engagement intensity.

Figure 3. Distribution of Last Login Gap Days

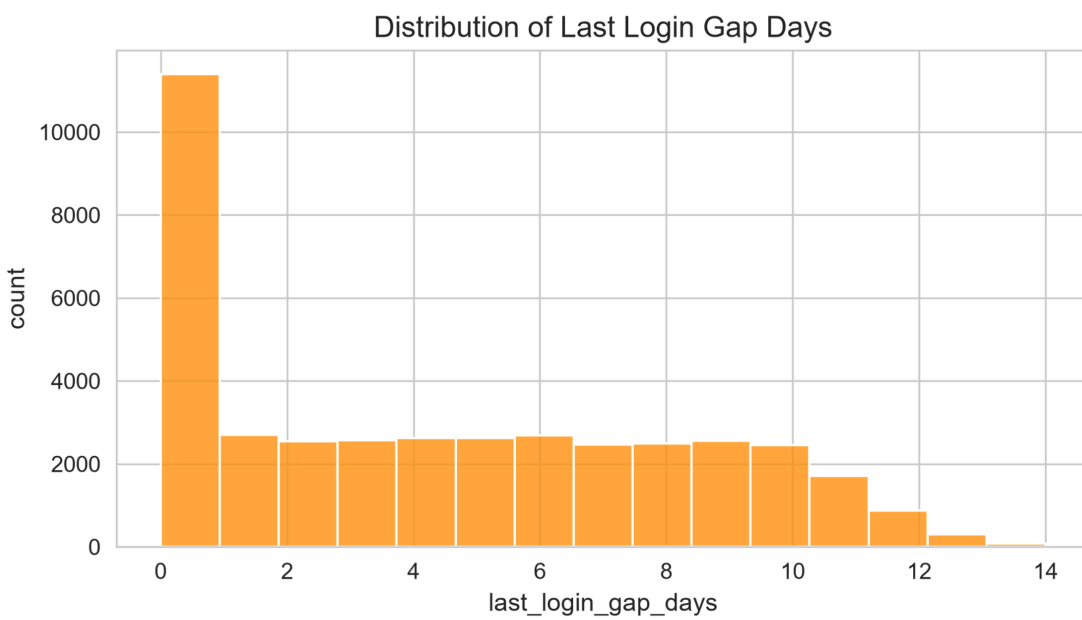

Figure 3 shows the spread of login recency across players in the synthetic dataset. Larger login gaps are associated with weaker recent engagement and higher modeled churn risk.

Table 2. Regional Churn-Risk Comparison

Region	Churn-Risk Rate
VN	22.9%
SG	22.3%
ID	22.2%
MY	22.1%
TH	22.7%
PH	22.2%

Figure 4. *Regional Comparison of Synthetic Churn-Risk Rates*

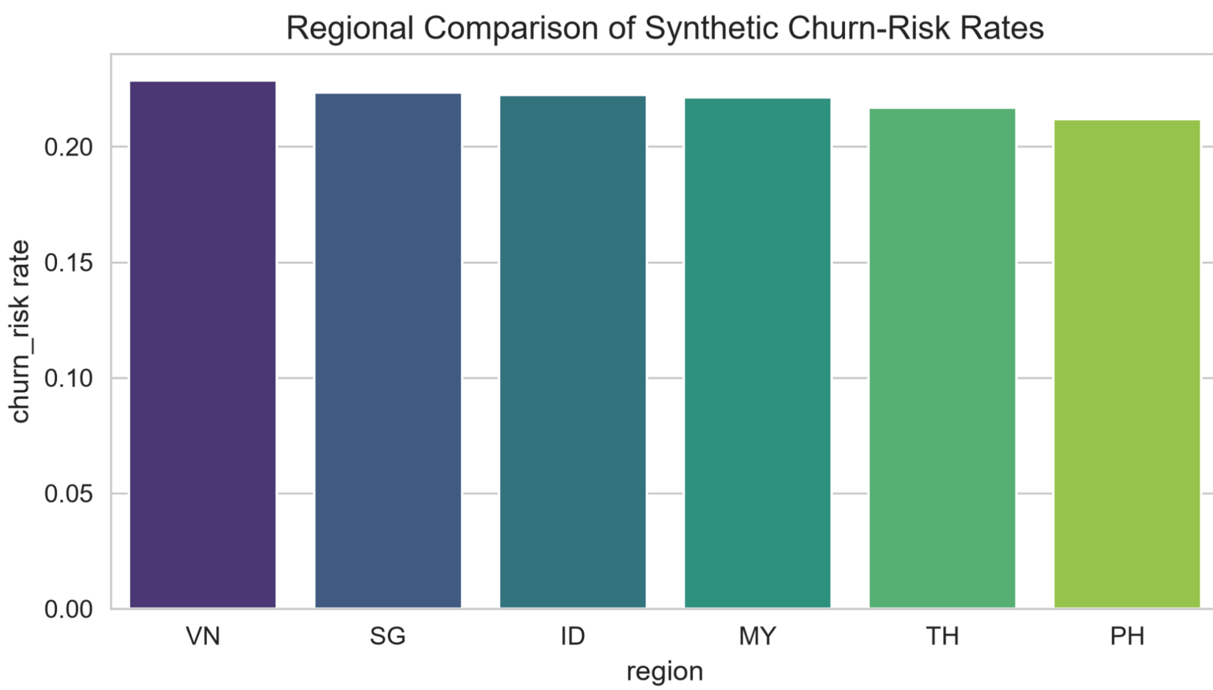

Figure 4 compares synthetic churn-risk rates across SEA-inspired player segments. Vietnam shows the highest modeled churn-risk rate at 22.9%, while the Philippines has the lowest at 21.2%. These values are synthetic and included for cohort-analysis realism rather than market inference.

6. Model Comparison Results

The cross-validation results show that all three tree-based models perform well in the synthetic churn-risk setting, but their strengths differ across metrics. Random Forest provides a solid baseline, though it is weaker than the two boosted-tree models in overall discrimination. XGBoost and LightGBM both outperform the baseline and are more suitable for this retention-risk use case.

Among the evaluated models, XGBoost delivered the strongest overall performance. It achieved 0.9927 AUC, 0.9637 accuracy, 0.9346 precision, 0.8979 recall, and 0.9158 F1-score. LightGBM was the second-best model overall, with 0.9926 AUC, 0.9543 accuracy, 0.8626 precision, 0.9429 recall, and 0.9009 F1-score. Random Forest followed with 0.9895 AUC, 0.9554 accuracy, 0.8774 precision, 0.9273 recall, and 0.9016 F1-score.

These results show a clear trade-off between overall balance and recall emphasis. XGBoost is the strongest model overall when considering AUC, precision, accuracy, and F1-score together. LightGBM remains notable because it produces the highest recall, which may still be attractive in a more intervention-heavy retention workflow where missing a higher-risk player is especially costly.

Table 3. Cross-Validation Model Comparison

Model	Accuracy	Precision	Recall	F1-score	AUC
XGBoost	0.9637	0.9346	0.8979	0.9158	0.9927
LightGBM	0.9543	0.8626	0.9429	0.9009	0.9926
Random Forest	0.9554	0.8774	0.9273	0.9016	0.9895

Figure 5. *Model Performance Comparison Across Key Metrics*

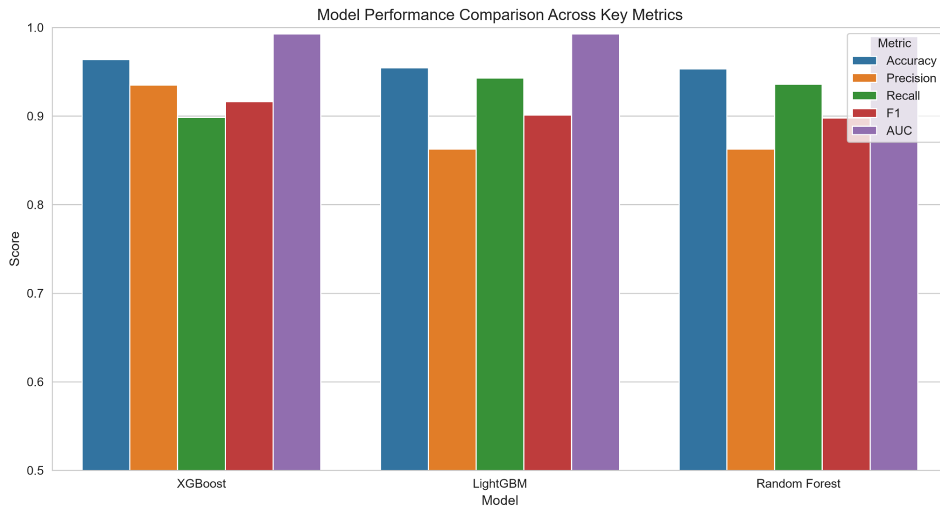

Figure 5 compares Random Forest, XGBoost, and LightGBM across accuracy, precision, recall, F1-score, and AUC. XGBoost achieves the strongest overall balance, while LightGBM delivers the highest recall.

7. Final Model Evaluation

The final evaluation focuses on XGBoost because it provides the strongest overall balance across the core evaluation metrics. Using cross-validated predictions, the model achieved 0.9637 accuracy, 0.9344 precision, 0.8979 recall, 0.9158 F1-score, and 0.9926 AUC. These results indicate that the model separates lower-risk and higher-risk players very effectively within the synthetic target framework.

The confusion matrix further supports this conclusion. The model correctly classified 30,671 lower-risk players and 7,909 higher-risk players. It produced 555 false positives and 899 false negatives. In practical terms, the false negatives are the more important error type because they represent higher-risk players who would not be surfaced for retention or reactivation intervention.

Class-specific results also remain strong. For the higher-risk class, XGBoost produced 0.9344 precision, 0.8979 recall, and 0.9158 F1-score. This suggests that the workflow is effective at identifying elevated churn risk while keeping misclassification at a manageable level. Although this is still a synthetic modeling exercise rather than a production system, the final evaluation demonstrates a robust end-to-end retention-risk workflow for portfolio presentation.

Table 4. Final XGBoost Metrics

Metric	Value
Accuracy	0.9637
Precision	0.9344
Recall	0.8979
F1-score	0.9158
AUC	0.9926

Table 5. Final Classification Summary

Class	Precision	Recall	F1-score	Support
Lower Risk	0.9715	0.9822	0.9768	31,226
Higher Risk	0.9344	0.8979	0.9158	8,808
Accuracy	0.9637			40,034

Table 6. Confusion Matrix Values

Actual \ Predicted	Lower Risk	Higher Risk
Lower Risk	30,671	555
Higher Risk	899	7,909

Figure 6. *Churn Risk by Recent Match Volume*

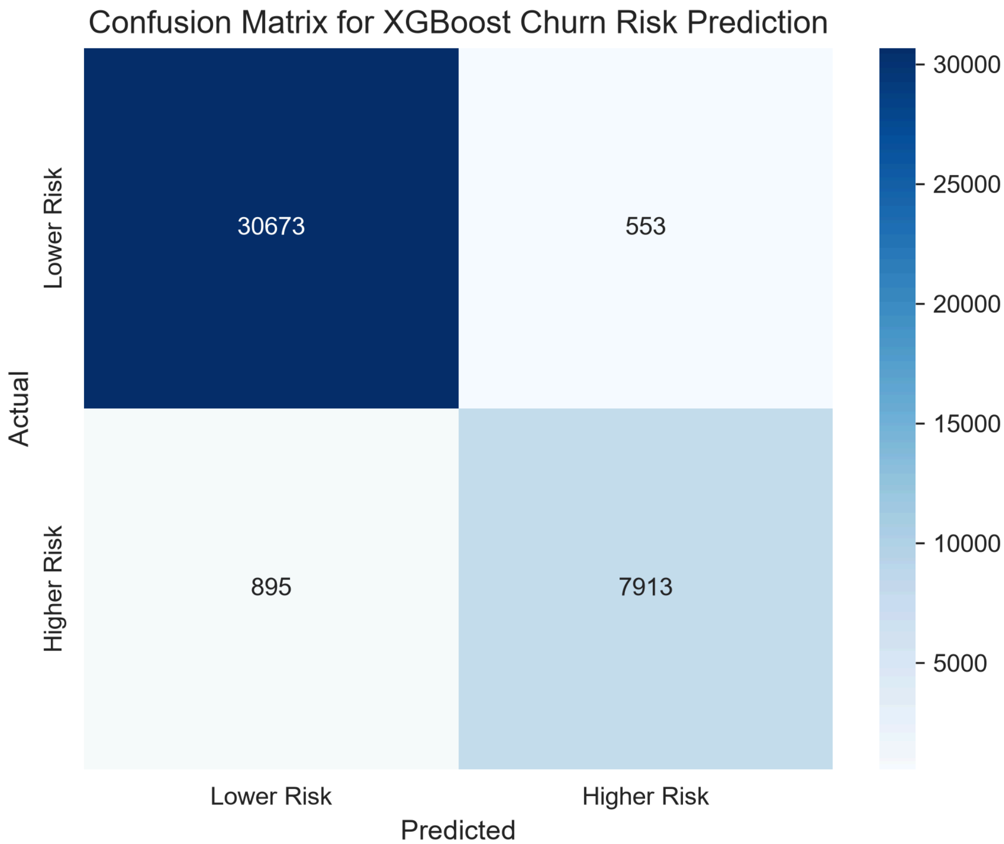

Figure 6 presents the confusion matrix for the final XGBoost workflow. The model correctly identifies 7,909 higher-risk players and 30,671 lower-risk players, with 899 false negatives and 555 false positives.

8. Churn Risk by Recent Match Volume

To better interpret behavioral risk patterns, churn risk was examined against recent match activity using match-volume buckets. The results show a steep decline in churn risk as recent play volume increases. Players with only 0 to 2 matches in the last 7 days have a churn-risk rate of 91.5%, while players with 3 to 5 matches show a risk rate of 37.7%.

The pattern becomes much more favorable for more active cohorts. Players with 6 to 10 matches in the last 7 days show a churn-risk rate of 6.9%, while players with 11 to 20 matches fall to just 0.5%. For players with more than 20 matches in the last 7 days, the synthetic churn-risk rate falls to 0.0% in the current generated dataset.

This pattern strongly reinforces the idea that recent participation is the most important behavioral signal in the synthetic retention-risk framework. From a business perspective, the chart is useful because it converts model logic into a directly interpretable operational pattern. It shows that declining recent activity is closely associated with elevated churn risk and can be used as a practical intervention signal even before more complex model outputs are examined.

Table 7. Churn Risk by Match-Volume Bucket

Matches in Last 7 Days	Risk Rate	Player Count
0-2	91.5%	7,278
3-5	37.7%	4,138
6-10	6.9%	7,616
11-20	0.5%	14,260
21+	0.0%	6,742

Figure 7. *Churn Risk by Recent Match Volume*

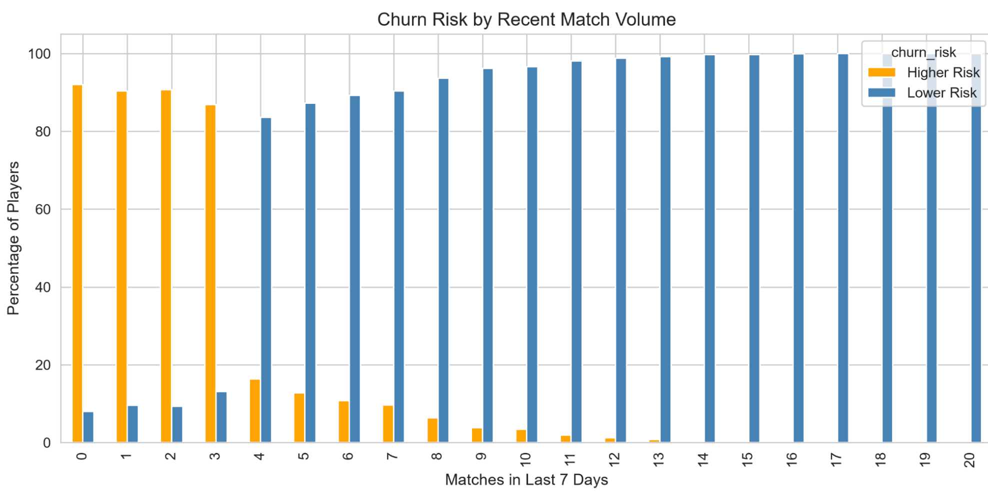

Figure 7 shows that churn risk declines sharply as recent match activity increases. Players with only 0 to 2 matches in the last 7 days have a modeled risk rate of 91.5%, compared with 37.7% for players with 3 to 5 matches, 6.9% for players with 6 to 10 matches, and 0.5% for players with 11 to 20 matches.

9. Model Explainability and Feature Importance

Model explainability was assessed through feature-importance analysis on the final XGBoost workflow. The most influential feature by a large margin is matches_last_7d, with an importance value of 0.5545. This is followed by active_days_last_7d at 0.1637, indicating that recent activity volume and recent activity consistency dominate the model's decision logic.

The next most important features are last_login_gap_days, spender_flag, matches_last_30d, report_count_30d, afk_count_30d, spend_tier_none, tutorial_completion, and progression_velocity_30d. Together, these results suggest that the model is learning a sensible risk structure. Players are more likely to be classified as higher risk when they show weaker recent participation, longer login gaps, no spending activity, lower tutorial completion, slower progression, or more friction-related signals such as reports and AFK counts.

From a portfolio perspective, these results are important because they show that the workflow is not only predictive, but also interpretable. The model is relying on intuitive lifecycle and engagement features rather than opaque or implausible patterns. This makes the project stronger as a demonstration of retention-risk reasoning in a gaming context.

Table 8. Top Feature Importance Drivers

Rank	Feature	Importance
1	matches_last_7d	0.5545
2	active_days_last_7d	0.1637
3	last_login_gap_days	0.0212
4	spender_flag	0.0210
5	matches_last_30d	0.0205
6	report_count_30d	0.0196
7	afk_count_30d	0.0192
8	spend_tier_none	0.0190
9	tutorial_completion	0.0134
10	progression_velocity_30d	0.0119

Figure 8. *Top Feature Importance Drivers for XGBoost Churn Risk Prediction*

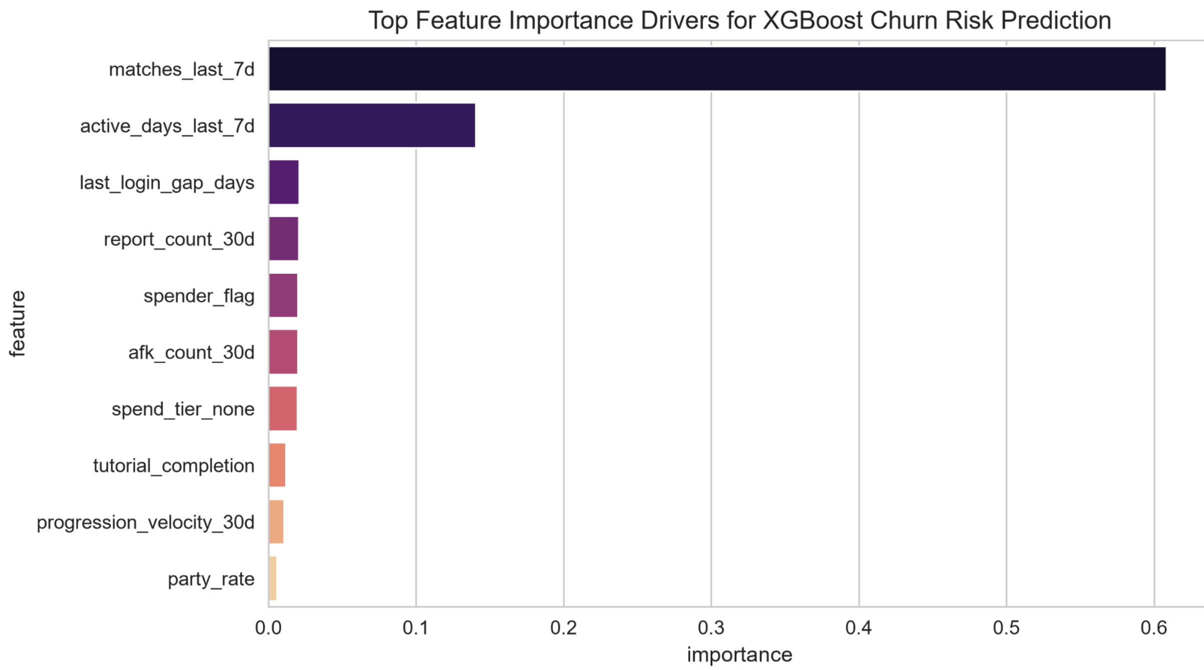

Figure 8 shows the most influential features in the final XGBoost model. Recent activity volume, recent activity consistency, and login recency dominate the ranking, followed by spending state, broader 30-day activity, and friction-related indicators.

10. Business Implications

This workflow is most directly applicable to early retention monitoring. Players with very low recent match volume, fewer active days, and longer login gaps can be identified before they fully disengage. In practice, that could support lower-cost interventions such as progression nudges, onboarding reinforcement, event prompts, or social-play recommendations.

A secondary use case is reactivation targeting. Players who still show some historical value, but now exhibit widening inactivity and weaker recent engagement, may be suitable candidates for targeted re-entry campaigns. In this setting, the model can help prioritize which cohorts should receive intervention first.

The results also suggest that a useful retention workflow should combine multiple signal types rather than rely on one engagement metric alone. Recent activity remains the strongest driver, but progression, spending state, tutorial completion, and friction indicators all add useful context for identifying elevated churn risk.

11. Limitations

This project has several important limitations. First, the dataset is synthetic and does not represent real Wild Rift production telemetry. Second, the target variable is engineered rather than observed from true churn outcomes. As a result, the results should not be interpreted as live operational performance.

Third, the feature relationships are designed to be realistic enough for portfolio demonstration, but they remain simulated. That means the workflow is useful for showing modeling structure, reasoning, and interpretation, but not for drawing conclusions about real player behavior or real market-level retention performance.

Finally, the model evaluation is entirely offline. A real deployment setting would require threshold tuning, intervention design, policy constraints, monitoring, and online validation against actual retention outcomes. Those steps are outside the scope of this project.

12. Conclusion

This project demonstrates a complete churn-risk modeling workflow using a synthetic dataset designed to approximate player lifecycle patterns in a fast-paced mobile MOBA environment. Among the evaluated models, XGBoost delivered the strongest overall performance, with 0.9927 AUC, 0.9637 accuracy, 0.9346 precision, 0.8979 recall, and 0.9158 F1-score in cross-validation.

The results suggest that recent activity, login recency, progression, spending state, and friction-related signals are all useful for identifying elevated short-term retention risk. Most importantly, the project shows how a public-facing portfolio artifact can demonstrate data design, preprocessing, model evaluation, feature interpretation, and business translation without exposing proprietary data.

In a real production setting, the next step would be threshold tuning, intervention design, and validation against observed retention outcomes. Within the current synthetic setup, however, the project already provides a strong portfolio example of applied churn-risk modeling in a Wild Rift-inspired environment.

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